How, with his severe disability, has Hawking been able to out-think and out-intuit his leading colleague-competitors, people like Roger Penrose, Werner Israel, and (as we shall see) Yakov Borisovich Zel'-dovich? They had the use of their hands; they could draw pictures and perform many-page-Iong calculations on paper—calculations in which one records many complex intermediate results along the way, and then goes back, picks them up one by one, and combines them to get a final result; calculations that I cannot conceive of anyone doing in his head. By the early 1970s, Hawking's hands were largely paralyzed; he could neither draw pictures nor write down equations. His research had to be done entirely in his head.

Because the loss of control over his hands was so gradual, Hawking has had plenty of time to adapt. He has gradually trained his mind to think in a manner different from the minds of other physicists: He thinks in new types of intuitive mental pictures and mental equations that, for him, have replaced paper-and-pen drawings and written equations. Hawking's mental pictures and mental equations have turned out to be more powerful, for some kinds of problems, than the old paper-and-pen ones, and less powerful for others, and he has gradually learned to concentrate on problems for which his new methods give greater power, a power that nobody else can begin to match.

Hawking's disability has helped him in other ways. As he himself has often commented, it has freed him from the responsibility of lecturing to university students, and he thus has had far more free time for research than his more healthy colleagues. More important, perhaps, his disease in some ways has improved his attitude toward life.